ILLUSTRATIVE EXEMPLAR — “what good looks like”

Predetermined Change Control Plan (PCCP) & GMLP

Autonomous AI diagnostic — diabetic retinopathy detection (Class II, De Novo)

Grounded in (public record): IDx-DR / LumineticsCore (Digital Diagnostics) — FDA De Novo DEN180001, April 2018, first autonomous AI diagnostic. Pivotal trial: 900 patients across 10 primary-care sites; 87.2% sensitivity, 90.7% specificity vs the Wisconsin Fundus Photograph Reading Center. Public sources: FDA De Novo DEN180001; University of Iowa / Digital Diagnostics announcements (2018).

Regulatory basis: FDA PCCP final guidance (2024); GMLP guiding principles (2021); FDA AI lifecycle draft (2025).

1. Device & intended use

Software-only autonomous AI that analyzes fundus images from a non-mydriatic camera at the point of care and returns a screening result for more-than-mild diabetic retinopathy (mtmDR), without specialist over-read. The reconstruction below illustrates how a PCCP would be structured for such a device under current guidance.

2. Description of modifications (bounded)

- Periodic **re-training** on additional labeled fundus images (same indication and population).
- **Operating-point** re-tuning within a pre-specified bound to maintain the sensitivity/specificity trade-off.
- Support for **additional compatible cameras** validated to the image-quality specification.
- **No change** to intended use, target population, or output categories.

3. Modification protocol

Step	Requirement
Data	≥ N new images; representative across age, sex, race/ethnicity, and disease severity; independent frozen test set; documented provenance
Reference standard	Reading-center grading (ETDRS) by certified graders, as in the pivotal study
Re-training	Versioned pipeline; no test-set leakage; change log
Acceptance gates	Sensitivity ≥ 85% and specificity ≥ 82.5% overall **and** within each subgroup; no regression vs prior version
Release	Versioned, signed model artifact with rollback; labeling/version note updated

4. Reference performance (pivotal, public)

Metric	Result	Reference
Sensitivity (mtmDR)	87.2%	Wisconsin Fundus Photograph Reading Center (ETDRS)
Specificity (mtmDR)	90.7%	Wisconsin Fundus Photograph Reading Center (ETDRS)

Metric	Result	Reference
Study	900 patients, 10 primary-care sites	Prospective pivotal trial (2018)

5. Impact assessment

Risks of the bounded changes — performance drift and subgroup degradation — are controlled by locked acceptance gates (including per-subgroup), independent test data, postmarket monitoring, and explicit halt/rollback criteria. Any change outside this envelope (e.g., new indication, new output) falls **outside the PCCP** and requires a new marketing submission.

6. GMLP evidence (10 principles)

#	Principle	Evidence (illustrative)
1	Multidisciplinary expertise	Clinical + ML + security across lifecycle
2	Good software & security practice	SDLC + security risk file
3	Representative data	Multi-site, demographically balanced cohort
4	Train/test independence	Frozen, audited test set
5	Reference standard	Reading-center ETDRS adjudication
6	Model tailored to use	Lesion-based pipeline for mtmDR
7	Human-AI team	Autonomous use validated for primary care
8	Clinically relevant testing	Prospective primary-care trial
9	Clear user information	Image-quality feedback + result labeling
10	Deployed monitoring	Real-world performance + drift triggers

7. Transparency

Each authorized release adds a dated version note (data summary, performance delta, affected configurations) to labeling so users understand what changed and the validated population.